

Differentialverstärker

$$I_c = -I_R$$

$$I_R = \frac{u_a(t)}{R} I_C = C * \frac{du_c(t)}{dt}$$

$$u_a(t) = -RC \cdot \frac{du_C(t)}{dt}$$

Integrator

$$I_R = -I_C$$

$$\frac{u_e(t)}{R} = -C \cdot \frac{du_a(t)}{dt}$$

$$\int_0^t \frac{du_a(t)}{dt} = - \int_0^t \frac{u_e(t)}{RC} dx$$